Management summary: Vacuum-salt plant with vapour recompression evaporator (US 5,366,514 flowsheet)

Status: DRAFT for engineering review, not approved. Generated by PlantAgent from the workspace record; every number carries its source.

Question: Prepare the data sheet for the new compressor K-202 from the last verified proposal, a work order for cleaning the main heater E-201, and a one-page summary for my manager.

Decision proposed: Clean heater, add 49.5 tonne compressor, cap purge steam

At 20% more feed the compressor needs 120% of suction and the fouled main heater 111.3% of area per run vrc_120_check. Cleaning E-201 to 2500 plus a 49.5 t/h parallel K-202 and holding purge steam at 20.9 t/h clears all checks at 89.0%, 90.0% and 88.3% per run vrc_120_K202_clean.

(Source: answer 6 (muse-spark-1.3-contributor).)

Capacity

Item	Value	Source
Target	120 % of today	cases/salt_mvr/case.json target_scale / answer 6 (muse-spark-1.3-contributor)
Capacity today	100.1 %	answer 6 (muse-spark-1.3-contributor) calculation record
Capacity with all verified changes	to confirm	answer 6 (muse-spark-1.3-contributor) combined verification
First limit after	to confirm	answer 6 (muse-spark-1.3-contributor) combined verification
Reaches target	no	answer 6 (muse-spark-1.3-contributor) combined verification

Changes and cost class

Change	Verified	Cost class	Capacity alone	Source
Clean main heater E-201	yes	<$100k	100.1 %	answer 6 (muse-spark-1.3-contributor)
Add parallel compressor K-202	no (change 1 combines path and add_unit; only add_unit plus reroute can be split)	$100k-$1M	—	answer 6 (muse-spark-1.3-contributor)
Hold purge steam at today's rate	yes	no capex	98.5 %	answer 6 (muse-spark-1.3-contributor)

Risks

Risk	Detail	Source
Relief on E-201 ST side	isolatable: relief 7002/PSV-202 only through 3003/C1: if one is closed the side has no relief	P&ID relief scan
Relief on E-301 ST side	isolatable: relief 7001/PSV-201 only through 3101/C1: if one is closed the side has no relief	P&ID relief scan

What needs approval (management of change)

Change	Part of	Source
V-201 evaporator heat-transfer coefficient: 2000 → 2500 W/m²K	Clean main heater E-201	answer 6 (muse-spark-1.3-contributor)
OVERHEAD_SPLIT branch fractions: [0.0843762616067824, 0.9156237383932176] → [0.0703, 0.9297]	Hold purge steam at today's rate	answer 6 (muse-spark-1.3-contributor)

Field checks before approval

Check	Where / how	Source
E-201 true clean U and current fouled U	Measure E-201 steam pressure/condensing T, brine inlet/outlet T and circulation flow at steady load to back-calculate U	answer 6 (muse-spark-1.3-contributor)
K-201/K-202 efficiency and discharge temperature	Check K-201 historian discharge T and power vs flow; confirm with compressor vendor for K-202	answer 6 (muse-spark-1.3-contributor)